

# SELECT from WORLD Tutorial

|           |                    |
|-----------|--------------------|
| Language: | English • 日本語 • 中文 |
|-----------|--------------------|

| name        | continent | area    | population | gdp          |
|-------------|-----------|---------|------------|--------------|
| Afghanistan | Asia      | 652230  | 25500100   | 20343000000  |
| Albania     | Europe    | 28748   | 2831741    | 12960000000  |
| Algeria     | Africa    | 2381741 | 37100000   | 188681000000 |
| Andorra     | Europe    | 468     | 78115      | 3712000000   |
| Angola      | Africa    | 1246700 | 20609294   | 100990000000 |
| ...         |           |         |            |              |

In this tutorial you will use the SELECT command on the table world:

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# 1. 🤪

## Introduction

---

Read the notes about this table. Observe the result of running this SQL command to show the name, continent and population of all countries.

```
SELECT name, continent, population FROM world
```

[Submit SQL](#)[restore default](#)

## Correct answer

| name                | continent     | population |
|---------------------|---------------|------------|
| Afghanistan         | Asia          | 25500100   |
| Albania             | Europe        | 2821977    |
| Algeria             | Africa        | 38700000   |
| Andorra             | Europe        | 76098      |
| Angola              | Africa        | 19183590   |
| Antigua and Barbuda | Caribbean     | 86295      |
| Argentina           | South America | 42669500   |
| Armenia             | Europe        | 2817400    |

# 2. 😊

## Large Countries

How to use WHERE to filter records. Show the name for the countries that have a population of at least 200 million. 200 million is 200000000, there are eight zeros.

```
SELECT name FROM world
WHERE population > 200000000
```

[Submit SQL](#)[restore default](#)

## Correct answer

| name |
|------|
|------|

|               |
|---------------|
| Brazil        |
| China         |
| India         |
| Indonesia     |
| United States |

### 3. 😊

## Per capita GDP

Give the name and the **per capita GDP** for those countries with a population of at least 200 million.

*HELP: How to calculate per capita GDP*

```
select name, gdp/population from world
where population >200000000
```

Submit SQL

restore default

## Correct answer

| name          | gdp/population |
|---------------|----------------|
| Brazil        | 11115.2648     |
| China         | 6121.7106      |
| India         | 1504.7931      |
| Indonesia     | 3482.0205      |
| United States | 51032.2945     |

SELECT from WORLD Tutorial - x

basilsqlzoo.net/wiki/SELECT\_from\_WORLD\_Tutorial#Per\_capita\_GDP

Per capita GDP [edit](#)

3.

Give the **name** and the **per capita GDP** for those countries with a **population** of at least 200 million.

HELP:How to calculate per capita GDP

Submit SQL

restore default

Wrong answer. Some of the values are incorrect.

| name      | gdp            |
|-----------|----------------|
| Brazil    | 2081235000000  |
| China     | 19373586000000 |
| India     | 3736882000000  |
| Indonesia | 1391778000000  |
| Nigeria   | 506601000000   |
| Pakistan  | 376533000000   |

0:00 / 2:19



4. 😊

## South America In millions

Show the name and population in millions for the countries of the continent 'South America'. Divide the population by 1000000 to get population in millions.

```
select name,population/1000000 from world
where continent like 'south america'
```

Submit SQL

restore default

|                                  |         |
|----------------------------------|---------|
| Guyana                           | 0.7849  |
| Paraguay                         | 6.7834  |
| Peru                             | 30.4751 |
| Saint Vincent and the Grenadines | 0.1090  |
| Suriname                         | 0.5342  |
| Uruguay                          | 3.2863  |
| Venezuela                        | 28.9461 |

# 5. 😊

## France, Germany, Italy

Show the name and population for France, Germany, Italy

```
select name , population from world
where name in ('France', 'germany', 'italy')
```

Submit SQL

restore default

## Correct answer

| name    | population |
|---------|------------|
| France  | 65906000   |
| Germany | 80716000   |
| Italy   | 60782668   |

6. 😊

## United

---

Show the countries which have a name that includes the word 'United'

```
select name from world
where name like '%united%'
```

[Submit SQL](#)[restore default](#)

## Correct answer

| name                 |
|----------------------|
| United Arab Emirates |
| United Kingdom       |
| United States        |

7. 😊

## Two ways to be big

---

Two ways to be big: A country is **big** if it has an area of more than 3 million sq km or it has a population of more than 250 million.

Show the countries that are big by area or big by population. Show name, population and area.

```
select name , population , area from world
where area>3000000 or population >250000000
```

Submit SQL

restore default

## Correct answer

| name           | population | area     |
|----------------|------------|----------|
| Australia      | 23545500   | 7692024  |
| Brazil         | 202794000  | 8515767  |
| Canada         | 35427524   | 9984670  |
| China          | 1365370000 | 9596961  |
| India          | 1246160000 | 3166414  |
| Indonesia      | 252164800  | 1904569  |
| Russia         | 146000000  | 17125242 |
| United Kingdom | 61000000   | 243647   |

8. 😊

## One or the other (but not both)

**Exclusive OR (XOR).** Show the countries that are big by area (more than 3 million) or big by population (more than 250 million) but not both. Show name, population and area.

- Australia has a big area but a small population, it should be **included**.
- Indonesia has a big population but a small area, it should be **included**.
- China has a big population **and** big area, it should be **excluded**.
- United Kingdom has a small population and a small area, it should be **excluded**.

```
select name , population, area from world
where area>3000000 xor population>250000000;
```

Submit SQL

restore default

Correct answer

| name      | population | area     |
|-----------|------------|----------|
| Australia | 23545500   | 7692024  |
| Brazil    | 202794000  | 8515767  |
| Canada    | 35427524   | 9984670  |
| Indonesia | 252164800  | 1904569  |
| Russia    | 146000000  | 17125242 |

9. 😊

Rounding

Show the name and population in millions and the GDP in billions for the countries of the continent 'South America'. Use the ROUND function to show the values to two decimal places.

For Americas show population in millions and GDP in billions both to 2 decimal places.

Millions and billions

Missing decimals

x

```
select name , round(population/1000000,2),round(gdp/1000000000,2) from world
where continent like 'south america'
```

Submit SQL

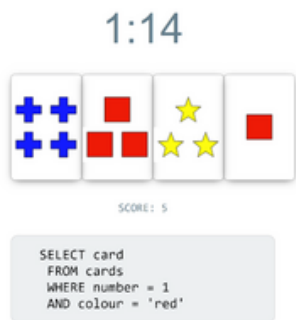
restore default

|           |       |        |
|-----------|-------|--------|
| Argentina | 42.67 | 477.03 |
| Bolivia   | 10.03 | 27.04  |



|                                  |       |        |
|----------------------------------|-------|--------|
| Peru                             | 30.48 | 204.68 |
| Saint Vincent and the Grenadines | 0.11  | 0.69   |
| Suriname                         | 0.53  | 5.01   |
| Uruguay                          | 3.29  | 49.92  |
| Venezuela                        | 28.95 | 382.42 |

Play the SELECT card game (<https://sqlzoo.net/40289347/>)



SELECT Card Game

# 10. 🤪

## Trillion dollar economies

Show the name and per-capita GDP for those countries with a GDP of at least one trillion (1000000000000; that is 12 zeros). Round this value to the nearest 1000.

**Show per-capita GDP for the trillion dollar countries to the nearest \$1000.**

```
select name , round(gdp/population,-3) from world
where gdp>1000000000000
```

Submit SQL

restore default

|                |       |
|----------------|-------|
| Canada         | 45000 |
| China          | 6000  |
| France         | 40000 |
| Germany        | 42000 |
| India          | 2000  |
| Italy          | 33000 |
| Japan          | 47000 |
| Mexico         | 10000 |
| Russia         | 14000 |
| South Korea    | 22000 |
| Spain          | 28000 |
| United Kingdom | 39000 |
| United States  | 51000 |



## Name and capital have the same length

Greece has capital Athens.

Each of the strings 'Greece', and 'Athens' has 6 characters.

**Show the name and capital where the name and the capital have the same number of characters.**

- You can use the LENGTH function to find the number of characters in a string

For Microsoft SQL Server the function LENGTH is LEN

```
SELECT name, capital
FROM world
WHERE length(name)=length(capital)
```

Submit SQL

restore default

## Correct answer

| name    | capital |
|---------|---------|
| Algeria | Algiers |

|          |          |
|----------|----------|
| Angola   | Luanda   |
| Armenia  | Yerevan  |
| Botswana | Gaborone |
| Canada   | Ottawa   |
| Djibouti | Djibouti |
| Egypt    | Cairo    |

## Variations

base

```
SELECT name, LENGTH(name), continent,  
LENGTH(continent), capital, LENGTH(capital)  
FROM world  
WHERE name LIKE 'G%'
```

mssql

```
SELECT name, LEN(name), continent,  
LEN(continent), capital, LEN(capital)  
FROM world  
WHERE name LIKE 'G%'
```

mysql

```
SELECT name, LENGTH(name), continent,  
LENGTH(continent), capital, LENGTH(capital)  
FROM world  
WHERE name LIKE 'G%'
```

oracle

```
SELECT name, LENGTH(name), continent,  
LENGTH(continent), capital, LENGTH(capital)  
FROM world  
WHERE name LIKE 'G%'
```

postgres

```
SELECT name, LENGTH(name), continent,  
LENGTH(continent), capital, LENGTH(capital)  
FROM world  
WHERE name LIKE 'G%'
```

12. 🤪

## Matching name and capital

The capital of Sweden is Stockholm. Both words start with the letter 'S'.

Show the name and the capital where the first letters of each match. Don't include countries where the name and the capital are the same word.

- You can use the function LEFT to isolate the first character.
- You can use <> as the **NOT EQUALS** operator.

```
SELECT name, capital  
FROM world  
where left(name,1) = left(capital,1) and name not like capital
```

Submit SQL

restore default

|                       |                           |
|-----------------------|---------------------------|
| Marshall Islands      | Majuro                    |
| Mexico                | Mexico City               |
| Monaco                | Monaco-Ville              |
| Mozambique            | Maputo                    |
| Niger                 | Niamey                    |
| Panama                | Panama City               |
| Papua New Guinea      | Port Moresby              |
| Sao Tomã and Prãncipe | Sãõ Tomã                  |
| South Korea           | Seoul                     |
| Sri Lanka             | Sri Jayawardenepura Kotte |
| Sweden                | Stockholm                 |
| Taiwan                | Taipei                    |
| Tunisia               | Tunis                     |



## All the vowels

**Equatorial Guinea** and **Dominican Republic** have all of the vowels (a e i o u) in the name. They don't count because they have more than one word in the name.

**Find the country that has all the vowels and no spaces in its name.**

- You can use the phrase `name NOT LIKE '%a%'` to exclude characters from your results.
- The query shown misses countries like Bahamas and Belarus because they contain at least one 'a'

```
SELECT name
FROM world
WHERE name like "%a%"
and name like '%e%'
and name like '%u%'
and name like '%i%'
and name like '%o%'
and name not like "% %"
```

Submit SQL

restore default

## Correct answer

name

Mozambique

## What Next

- BBC QUIZ
- You can to continue practising the the same techniques and gain more experience of the basic skills on the Nobel table. The `WHERE` statement using the `nobel` table.
- You can learn about nested statements, these are instructive and entertaining, but not essential for beginners. Nested `SELECT` statements using the `world` table.

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```
query.sql

SELECT name
FROM learners
WHERE course = 'SQL';
```